Fair Test · Staff Answers

Learning goal: I can plan a fair test by naming what to change, what to measure, and what to keep the same.

Arrival Answers

1. Box 1 shows a fair test.
2. One.
3. Control variables.
4. Yes. You can ask or pass.

Starter & Hinge Answers

Starter: A. If drop height also changed, we would not know whether size or height changed the fall time.

Hinge: A. Ramp height is what we change on purpose (IV); rolling distance is what we measure (DV).

Evidence Desk Cards

Parachute canopy sizes: 10x10, 20x20, 30x30 cm: Independent variable: we change canopy size on purpose.

Fall time to the floor in seconds (s): Dependent variable: we measure fall time in seconds.

Drop height fixed at 1.50 m above the floor: Control variable: every drop starts from the exact same height.

Same 10 g modelling-clay load on every canopy: Control variable: keeps gravity pull (weight) constant.

Same four 25 cm cotton strings and plastic sheet:

Control variable: keeps material and string length constant.

Using a heavier load on the largest parachute: Unfair test flaw: changing two variables makes the comparison invalid.

Check & Repair Answer

Changing more than one variable makes it impossible to tell which factor caused the change in fall time.

Exit Ticket Answers

1. Parachute canopy area (size). 2. Fall time in seconds (s). 3. Same drop height (1.5 m) and same load mass (10 g), so any change in fall time is caused only by canopy area.